

At the room like this, we keep everything the same. We raise the ceiling height. Easiest thing of all. We have a room just like this. We alter the windows or the natural lighting in some way. We have a room just like this. We make that wall curved, not so easy. Okay. So, we change the texture on the walls. Now, those are things that you could do quite easily, unsure, in unity. Or in unreal. Yeah. And, uh, but do you have to become familiar with unreal unity in order to do that? As as, like, the user? Yeah. Um... Not the viewer, as the person who's gonna change the room. Here, these students, they're gonna have two weeks. Mm hmm. in which to take... They've designed an experiment. And they say, Okay, I wanna see if room height affects concentration. Okay. Rooms that are higher than 10 feet are supposed to be better for, uh, you know, a creative thinking, and, uh, called divergent, and, uh, the thinking, and rooms less than 10 feet, more like 8 feet, are better for concentration. If you have to do multiplying, or type, you know, proofreading, or something that requires concentration, you're better in a little, in a lower room. And you can say, Well, how robust does that find? How many people have found that? So, undergraduate might come into the class and say, Okay, I wanna try that one. I said, Well, okay, it's been done before. and say, that's fine. It needs to be done, because the system, my system, will tell them, Hey, you know what? That's a really significant finding. We don't know how true that is. Hasn't been replicated enough. And so, they do it, and they might change the measuring devices we're using on people. They might change the tests, so it wouldn't be identical to the same one. That would be acceptable in that class. But they're gonna have to have a stimulation. They got two weeks in which to make the stimulate. Right? They got two weeks in which they're gonna set it up so that the task can be performed in that room. Because that's what they're supposed to do. So they're gonna have to figure out, how can I give somebody a task where they have to, like, do proofreading inside the room? Okay. And now, so that's why we're trying to have it parametric, so we can change things in it, like, Okay, I'm gonna just change a few things like this. I'm going to change some things like that. Or, here, here are some major changes. We made the curvy room. Here's the one room, the regular room, the regular room. Here's the curvy room. You can change some parameters on the curvy room. And maybe, I'm not saying it's wrong, but you can't do it in unity. That parametric models were the way that one wants to go. I think it should be doable, but I think we need to know which type, is it gonna be, like, behavioral experiment, or is it gonna be, like, actual experiment that's, like, physics space heavy? Well, um, so... There's no real physi... If somebody wanted to do a navigational study, Yeah. They wanted to know whether this kind of configuration is easier to remember than that kind of configuration. So that afterward, we could bring them back, so they could walk, they have to walk through the space. And then afterward, they come back to the beginnings, and one possibility would say, Okay, now. That room over there. Maybe we'll flash the room. Mm hmm. Point to where it is. And they'll have to have some way, so then we have to have some controls over what it is that they can do, but I presume we can get there. So they would then have to point. And now we count how far off the pointing it is for the navigation. I am inventing an experiment, and that would be more involved. They had to walk through the space, right? You could have memory tests, and you could have, like, navigation... It's a kind of memory text, but it's computing. Yeah. So they could do that. That would be the most physics kind of thing that would have to do. They're not gonna be, like, actually cooking in a space. Okay. It's not what they're gonna be doing. To think creating the rooms could be possible by the end of the summer, but the interactions, they have two weeks to make inside the rooms. I just don't know how

undergrad is. Well, most of them are not gonna be doing it for this particular thing. They're going to be having some activity. They'll put, like, the task, the interaction, task would be the test. I mean, what they have to do, and that what they have to do will be some cognitive task or something, or, we can let them walk through, and then afterward, we do these things where, okay, you know, we get them to do, it's called adaptive preference. You say, okay, and you're... The simplest one is like you prefer, A to B, A to B, A to B, A, to B, but just figuring out that you've got 20 of them, and in about 8 questions, you can smartly compare them all. But it does it in a smart way. But there could be other questions, too, that are more interesting. Don't make me think of them. Uh, but, uh... how lighting effects, like, safety or something. Well, you know, like the point I'm making is, they could have the time in the room doing something. Tell them. Oh, that plays, you know, not hide and seek, but, you know, find something. Or look at something. You could give them tasks that make them look at the room. But afterward, you have to figure out what happened and what got into their head, and so afterward, you do the tests. They not in the room any longer. So they do the task in the room, and then we see if it's, you know, if it's a memory test, I don't need the room anymore. Right? If it was the pointing room, pointing thing, you can keep the broom constant, but I'm not going anywhere. I am just over here, like that. So I'll figure out, you know, 25 or 50 of those. things for them, and they'll be able to sort of pick it and motivate why they want that one. But it all starts with this idea that they can have a control condition, and manipulation condition, a treatment condition. And we're back to the point that we want a pretty realistic room. and we wanna be able to change it. If you can do that in unity, I have no special care, as long as the front end to unity is such that we can do it. Okay. As long as the front is in unity, I think so. Do you mind if it's gonna be, like, additional, like, the user has to install unity to use? No, I don't care. I mean, how much is it worth? Uh... I don't know. People wouldn't be able to do it off a Mac. Well, they would, but not the cheap Mac. Yeah. Macs are really good. They the best ones right now, in fact. Good for you. So, how much do you need to have to run something or run unicorn? Well, wait a second. There's the rendering. There is the making of it in unity. versus the running of it in unity. I think normal max, probably if I don't be able to run it. Do you have to have, like, top of the line mags? I'm not sure about, like, the new new Megs. Yeah. But they're gonna be pretty heavy to around. So, to run, you mean to, to, to... Well, we can give them the experimental thing to run it, but to make it as expensive, is... To making sure... Where'd you make it? I have my own PC. I have a pretty strong PC. Yeah. So, I don't have a Mac or, like, a strong enough neck to test. So I don't know how meg... Don worry about it. Just, just, let's just address the main question. Do you have to run the whole unity system to be able to design in unity? Yes. So is it even more intensive than manipulating it, or is it just, it may not be more intensive, 'cause you're doing something different. You're not, like, right? Not rendering things like crazy. And what helps to create, you know, the rooms, different ways, host them all on VR chat, and then the user will just teleport the test subject from room to room, and each instance will have the parametric change. Don't be a slider, but like you'll teleport them into a room with a high ceiling. And then you conduct the test here, then you'll be like test ended, you teleport them to another room with a low ceiling. Yeah, yeah, I didn't think that was a challenge. I thought the challenge was making the room right now. The actual running of the experiment could be done many ways. We wanted to use uni. to host it on GitHub pages. I have 2 or 3 VR worlds on GitHub pages, but the Fidelity is like polygon or something. So, you know, this is the problem. We have. We need to have fairly The room has to be realistic

enough so that you get so that it gives you a reliable experience. So, I'd say, 16? Oh, 12, 16. So you get a reliable experience of being in the room. If it's a crappy old paper room, you know, it's like a paper model. You can use paper model. I mean, architects use, but they're architects. They're looking at things. They're not, like, getting the same experience. So... But the basic question was, how easy, assuming engines were not the issue? How easy is it to modify the room inside VR? inside unity? As a user, or, like, creating the tool? Not creating the tool. They're making the stimuli. How easy is it to say, okay, make the ceiling high? It should be easier than what were you doing? Yeah. And why is that? Because, uh, infinity jam is a lot of function and infinity doesn't really work, and in order to get some of them work, we have to really take apart infinity, and but nobody's tried to do that yet, right? Yeah, he's a good apart, pretty dang hard. I had a whole app that ripped unity apart, and then from the data, and the dot G in or something like that. Yeah. which has the parameters, then when we got all the data from the one app, took it apart, and then from this data, we made the Gueys and version 4. Yeah, but the room. The room. That infinity gave you. Yeah. Take a parting to the check. Yeah. Well, the room and finishing gave you was terrible. The doors are, yeah. Yeah, I mean, you know, the doors, the windows, the... My temple, I think my temple turned out really good. I'm glad your temple turned out good. I don't think it looks like a temple. Okay. It's like the flagship, is it worship, temple, or something? Yeah, you know, I'm happy that you're happy. It's all I'll say. It should have gone way harder on it, but he said to stop against it. Well, we're not doing that one particular, you know? An experiment. The reason is, and I don't know why I'm dwelling on this, an experiment has got to be controlled. And you had so many changes in your temple with different size columns and different curves and domes and stuff. Those are all too many manipulations. Well, I, my, my, for that attempt, I set the bar, I said, Try all these things. So if I can do all these crazy ones, then I can definitely be easier. Yeah, but I would ask you, show me something that's plausible, and you still show me crazy ones. So I... Yeah, just imagine... I don't want to imagine. Did you get any answer from your thing yet? Yeah. Yeah. What's it say? It's the end of a quarter. I don't know. You know, Laura's fine, convinced. Can I propose to me, too? Oh, we do? If you take us into your lab as research assistance, we can get this, you know, taken to the next level over the summer. I have to pay you? No. Oh. I don't have money for this. I'm gonna pay you. I'm gonna pay things, I'm gonna pay things I know I'm gonna get. So you forgive me. I have great confidence in you. Just... Um, sure, I can do all of that. I'm happy to say that. You can say, you know, I'm delighted, in fact. I should let me rephrase all of this. What an opportunity. That is fantastic. I'm 100% in. Uh, no, I would, of course, but, uh, I think then maybe we're, I wanna think... I wanna review the possibilities. We could have two... So Skunkworks is an old term. They used to have this, like, in the military, and in other cases, in design places, where, you know, you got to invent things, and they would have... multiple buildings, where the teams, different teams would be working on the same problem. And they're all competing. They're not talking to each other. And so that... So, I'm happy to have the unity team. And I'm happy to have the infinitine team. And I'm even happy if somebody actually knows something about unity, or A frame, or unreal. I'm hoping I just came back from a conference, workshop, and somebody who works at the Max Plank Institute, and Germany, has gorgeous VR rooms. Gorgeous. that she uses, and I think she will be happy to share them. Suddenly, I'm gonna have VR rooms. Now, they, you know, she has, she manipulated them so that she was comparing them with, like, Imagine a room for coming in for medical treatment, and you, where you come to the front desk

to think, but it's a nice hospital, okay? So you come into that space, it looks nice. And she changed the curvy bit on the thing, one wood and the other was white, regular plaster. But she was an architect, and she's done her PhD in neuroscience on this stuff. So she's a smart woman, and she's no youngster, 'cause she was working for 10 years as an architect. So she probably made those VR things in the way architects make. Three dimensional models. And so she probably made them, and that's why they're good. And she probably made them in whatever she had to make them in. I'll ask her. So, I guess I'll ask her how they would be changed. So that would be possibly a third group, or, you know... Anyway, I'm just sharing. One group I would like to see if we can take infinitin further. If it turns out that we always get crappy doors, no matter what we try to do, and the lighting is always good. So we got from Infinigen to Unity. Yeah. Because you guys said, well, you're never going to get lighting out of finaging. And so then, why can't Unity take a 3D model? and apply lighting to the 3D model? Why does it have to be built inside unity? We can implement infinigen in it, but we just don't know how... No, you implementing a blender 3D model. It does come from infinity. Yeah. Yeah, but who cares? You just... By the time it gets to blender, it's just a 3D model, right? Yes. And it's, I don't know how blender defines those light surfaces. Does it define them in a way that captures the fact that they were generated light, or does it just capture them as, that's what the texture is? Depend on how we compress the bottle. Okay? Here's a foul size matter, uh, because out of the Boston, Tenage and Clouds are, like, half a gigabyte, just for one day. And we want to have our server running, so we have to compress it pretty hard. So after compressing it, the texture are a little bit messed up. That's obviously, like, a little bit clipping on the lighting system or even on the walls. So, are you, did you get good answers? We sent you the e mail. Okay, is it optimistic? Is it promising? We got it at credits, but we have the prawns, and the handoff, alcuments. Okay. MLG, the zip pile, and the classics were created. Okay, and you'll send me the prompt for cloud design. Yep, that's in there. Okay. Okay, great. Super. I really look forward to that. Oh. You never showed me. Are you sure you're out? I could send it to you, too. It's a part of my book. Okay, um, I kind of want to see it. Well, all right, we're gonna not solve anything right now, so... Okay, don't worry, y'all get your good grades. Um, but, um, you know, okay, and if you really want it, email me, and I'll say that you are one. But then you come in as you get a course credit for it, you know? I mean, if you come in, you get a... you have to pay? If you get course credit, you have to pay? Cons 199. Yeah. Yeah, I know what it is. Excuse me, I didn't mean it like that. I mean, I know that it's a 199. What I'm asking is, you're not enrolled in the summer, right? If you take a course, it's a credit. Yeah. Now, I could give you a 199 for the fall. You do it in the summer, you get a 199 a fall. better, yeah. That won't cost you anything. in liberal money. Cost you hours. Well, I get a little money. And how many days, dude? You told us that it was free at 12:30, Is that the case? Yeah, okay. Cool, cool. Sorry. We Okay, so we're all gonna email her. Let's do it. Uh... Okay... Let just do some resting. Okay, so email me if you're interested, okay? Okay, thank you so much, thank you.